

Mehul Agarwal

Software Engineer | Python & C++ | Full-Stack & Backend Development

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SUMMARY

Computer Science graduate (2026) with hands-on experience developing and maintaining software applications in Python and C++ across data engineering, backend, and full-stack systems. Skilled in collaborating within cross-functional, Agile teams to design solutions that meet customer requirements, troubleshoot and debug production issues, and uphold coding standards through consistent code review. Comfortable scoping ambiguous problems alongside technical and non-technical stakeholders and translating complex systems into solutions teammates can own; 700+ DSA problems solved.

EDUCATION

Jaypee Institute of Information Technology

Bachelor of Technology, Computer Science and Engineering

CGPA: 8.2 / 10

2022 – 2026

Noida, India

WORK EXPERIENCE

Navikenz

Mar 2026 – Jun 2026

Software Engineer Intern, Data Engineering and Backend

Noida, India

Developed and maintained ETL ingestion pipelines in Python on Microsoft Fabric, consolidating data from **3 heterogeneous sources** (Snowflake, Amazon S3, SharePoint) into unified target databases, cutting manual data-transfer effort by **60%**.

Designed source-to-target mappings and Apache Spark transformations across **5+ schemas**, applying validation and data-quality checks that reduced ingestion errors by **30%** and ensured production-grade reliability.

Collaborated with cross-functional engineers and non-technical product stakeholders in an Agile environment, troubleshooting and debugging pipeline failures to resolve **100%** of reported data-flow incidents within agreed SLAs.

Pristine Business Solutions

May 2025 – Jul 2025

Software Developer Intern, Enterprise Systems and Data Validation

Noida, India

Developed automated data validation pipelines in Python that improved workflow reliability by **35%** across enterprise production systems.

Debugged and optimized complex SQL queries through index tuning and query restructuring, reducing average execution time by **40%** on large datasets.

Built anomaly-detection monitoring and automated reporting tools, participated in code reviews, then documented and handed off tools to operations staff, increasing data accuracy and speeding decisions.

PROJECTS

NaviformatIQ – AI Multi-Agent Document Reformatting Platform

GitHub Live

Designed, developed, and deployed end-to-end a Python-based multi-agent software platform with **3 domain-agnostic workflows** (document generation, style transfer, compliance checking) using LangGraph and Azure OpenAI, meeting functional requirements gathered from prospective users.

Implemented a human-in-the-loop review flow plus a context-aware chat agent so non-technical users stay in control of every agent decision and can verify output before export.

Built a FastAPI backend orchestrating multi-agent pipelines with Neon Postgres, Qdrant vector search, and Server-Sent Events for real-time streaming of agent progress; React (Vite + Tailwind) frontend exports final reports to .docx and .pdf.

Tech Stack: Python, FastAPI, LangGraph, RAG, Azure OpenAI, React, Docker, Qdrant, PostgreSQL

Sovaria – Authority-Driven Backend-as-a-Service Platform

GitHub Live

Designed and implemented end-to-end a distributed Backend-as-a-Service system in Python and TypeScript spanning **11 specialized engines** (identity, policy, NoSQL/SQL storage, temporal versioning, realtime, simulation, functions, billing) behind a single Identity → Policy → Execution pipeline.

Delivered low-latency WebSocket realtime (5–20 ms from commit) and shipped identical Python and TypeScript SDKs plus a CLI with dry-run simulation for CI/CD, enabling adoption without reading the internals.

Tech Stack: Python, TypeScript, Distributed Systems, NoSQL and SQL Engines, WebSockets, REST API

Spell Check and Correction System

GitHub

Built a C++ spell checker that detects misspelled words against a dictionary file and generates ranked candidate corrections using the Levenshtein distance algorithm.

Implemented edit-distance computation from scratch to score every dictionary word against a misspelled input, then sorted and surfaced the closest matches to the user.

Tech Stack: C++, Data Structures, Algorithms

LEADERSHIP AND COMMUNITY

Mentored junior students in Data Structures and Algorithms (Python and C++), breaking down problem-solving patterns and guiding hands-on practice – translating technical concepts for less-experienced peers.

Lead Coordinator, Optica Club (2025): scoped and ran a campus coding contest end-to-end – problem selection, logistics, and prize-money distribution – for student participants.

SKILLS

Programming Languages: Python, C++, TypeScript, SQL, JavaScript

Software Development Practices: Agile Methodologies, Code Review, Debugging, Unit Testing, CI/CD, Problem-Solving & Analytical Skills

AI/ML and GenAI: LLMs, LangChain, LangGraph, RAG, MCP, Azure OpenAI, Vector Databases, TensorFlow

Web and Backend: FastAPI, Node.js, React.js, REST APIs, WebSockets, Server-Sent Events

Data Engineering: ETL/ELT Pipelines, Apache Spark, Microsoft Fabric, Snowflake, Data Modeling, Pandas, NumPy

Cloud, DevOps & Databases: Microsoft Azure, AWS (S3), Docker, Git, GitHub, PostgreSQL, MySQL, MongoDB, Qdrant